HDOC DAY – 64

Algorithm

1. Start.
2. Read the size of the array.
3. Declare the variable-length array nums[size].
4. Read all the elements of the array.
5. Pass the address of nums[] and size to duplicate_element().
6. Initialize result = 0.
7. Traverse the array once.
8. XOR result with each array element.
9. XOR result with the numbers from 1 to size - 1.
10. After the traversal, result contains the repeated element.
11. Display the repeated element.
12. Stop.

C PROGRAM:

```
#include <stdio.h>


void duplicate_element(int *nums, int size)
{
   int i;
   int result = 0;


   for (i = 0; i < size; i++)
   {
      result = result ^ *(nums + i);


      if (i < size - 1)
```

```
        result = result ^ (i + 1);
    }

    printf("Repeated element: %d\n", result);
}

int main()
{
    int size, i;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter the array elements: ");
    for (i = 0; i < size; i++)
    {
        scanf("%d", &nums[i]);
    }

    duplicate_element(nums, size);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Size	Array	Expected Output

1	5	1 3 4 2 2	Repeated element: 2
2	5	3 1 3 4 2	Repeated element: 3
3	6	1 4 5 2 3 4	Repeated element: 4
4	4	1 2 3 1	Repeated element: 1
5	6	1 2 3 4 5 5	Repeated element: 5